

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.51_2)

Metric

$$ds^2 = -dt \otimes dt + \left(-\frac{F(t)^2}{r^2 - 1} \right) dr \otimes dr + V(t, z, r)^2 dz \otimes dz + r^2 F(t)^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, z, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\frac{\sqrt{\frac{1}{2}} F(t)}{\sqrt{-r^2 + 1}} \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\frac{\sqrt{\frac{1}{2}} F(t)}{\sqrt{-r^2 + 1}} \right) dr$$

$$m = \left(\sqrt{\frac{1}{2}} V(t, z, r, \phi) \right) dz + \left(i \sqrt{\frac{1}{2}} r F(t) \right) d\phi$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} V(t, z, r, \phi) \right) dz + \left(-i \sqrt{\frac{1}{2}} r F(t) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2} \sqrt{|r+1|} \sqrt{|r-1|}}{2 F(t)} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2} r^2}{2 F(t) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{2}}{2 F(t) \sqrt{|r+1|} \sqrt{|r-1|}} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2 V(t, z, r, \phi)} \right) dz + \left(\frac{i \sqrt{2}}{2 r F(t)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 V(t, z, r, \phi)} \right) dz + \left(-\frac{i \sqrt{2}}{2 r F(t)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2} \sqrt{|r+1|} \sqrt{|r-1|}}{2 F(t)} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2} r^2}{2 F(t) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{2}}{2 F(t) \sqrt{|r+1|} \sqrt{|r-1|}} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2 V(t, z, r, \phi)} \partial_z + \frac{i \sqrt{2}}{2 r F(t)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 V(t, z, r, \phi)} \partial_z - \frac{i \sqrt{2}}{2 r F(t)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)} + \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\sqrt{2} r}{4 F(t) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{2} \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{2}}{4 r F(t) \sqrt{|r+1|} \sqrt{|r-1|}}$$

$$\sigma = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t, z, r, \phi)}{4V(t, z, r, \phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial r}V(t, z, r, \phi)}{4F(t)V(t, z, r, \phi)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}r}{4F(t)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}\frac{\partial}{\partial r}V(t, z, r, \phi)}{4F(t)V(t, z, r, \phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}}{4rF(t)\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t, z, r, \phi)}{4V(t, z, r, \phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial r}V(t, z, r, \phi)}{4F(t)V(t, z, r, \phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}r}{4F(t)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}\frac{\partial}{\partial r}V(t, z, r, \phi)}{4F(t)V(t, z, r, \phi)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}}{4rF(t)\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\lambda = \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t, z, r, \phi)}{4V(t, z, r, \phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial r}V(t, z, r, \phi)}{4F(t)V(t, z, r, \phi)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}r}{4F(t)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}\frac{\partial}{\partial r}V(t, z, r, \phi)}{4F(t)V(t, z, r, \phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}}{4rF(t)\sqrt{|r+1|}\sqrt{|r-1|}}$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{2}\frac{\partial}{\partial \phi}V(t, z, r, \phi)}{4rF(t)V(t, z, r, \phi)}$$

$$\beta = \frac{i\sqrt{2}\frac{\partial}{\partial \phi}V(t, z, r, \phi)}{4rF(t)V(t, z, r, \phi)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{4F(t)}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{8F(t)} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)}{8F(t)|r+1||r-1|} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{8F(t)|r+1||r-1|}$$

Ricci Tensor Components

$$\begin{aligned} \Phi_{00} = & \frac{r^4 \frac{\partial}{\partial t} F(t)^2}{8(r^4 - r^2)F(t)^2} - \frac{r^2|r+1||r-1|\frac{\partial}{\partial t} F(t)^2}{8(r^4 - r^2)F(t)^2} - \frac{r^4 \frac{\partial^2}{(\partial t)^2} F(t)}{4(r^4 - r^2)F(t)} + \frac{r^4 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4(r^4 - r^2)F(t)V(t, z, r, \phi)} - \frac{r^4 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8(r^4 - r^2)V(t, z, r, \phi)} + \frac{r^2|r+1||r-1|\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8(r^4 - r^2)V(t, z, r, \phi)} \\ & + \frac{r^4 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2(r^4 - r^2)F(t)V(t, z, r, \phi)} - \frac{r^4 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{r^6 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{r^4|r+1||r-1|\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} \\ & + \frac{r^5 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{r^4}{8(r^4 - r^2)F(t)^2} - \frac{r^2|r+1||r-1|}{8(r^4 - r^2)F(t)^2} - \frac{r^2 \frac{\partial}{\partial t} F(t)^2}{8(r^4 - r^2)F(t)^2} + \frac{r^2 \frac{\partial^2}{(\partial t)^2} F(t)}{4(r^4 - r^2)F(t)} - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4(r^4 - r^2)F(t)V(t, z, r, \phi)} \\ & + \frac{r^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8(r^4 - r^2)V(t, z, r, \phi)} - \frac{r^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2(r^4 - r^2)F(t)V(t, z, r, \phi)} + \frac{r^2 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} \\ & + \frac{r^2|r+1||r-1|\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{3r^3 \frac{\partial}{\partial r} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{r|r+1||r-1|\frac{\partial}{\partial r} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{r^2}{8(r^4 - r^2)F(t)^2} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} \\ & - \frac{r^2 \frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{|r+1||r-1|\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} \end{aligned}$$

$$\Phi_{01} = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4rF(t)V(t, z, r, \phi)} - \frac{i \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rF(t)^2 V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4rF(t)^2 V(t, z, r, \phi)} + \frac{i \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\Phi_{02} = -\frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} - \frac{1}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}$$

$$\Phi_{10} = -\frac{i}{4rF(t)}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) + \frac{i\sqrt{|r+1|}\sqrt{|r-1|}}{4rF(t)^2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) + \frac{i}{4rF(t)^2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) - \frac{i\sqrt{|r+1|}\sqrt{|r-1|}}{4r^2F(t)^2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)$$

$$\begin{aligned} \Phi_{11} = & -\frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi)} \\ & - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} \\ & - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi)} \end{aligned}$$

$$\begin{aligned}\Phi_{12} = & \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)} - \frac{i r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} \\ & + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}}\end{aligned}$$

$$\Phi_{20} = -\frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} - \frac{1}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}$$

$$\begin{aligned}\Phi_{21} = & -\frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)} + \frac{i r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} \\ & - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}}\end{aligned}$$

$$\begin{aligned}\Phi_{22} = & \frac{\frac{\partial}{\partial t} F(t)^2}{8 F(t)^2} - \frac{r^2 \frac{\partial}{\partial t} F(t)^2}{8 F(t)^2 |r+1| |r-1|} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1| |r-1|} + \frac{r^2 \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\ & - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} - \frac{r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{1}{8 F(t)^2} - \frac{r^2}{8 F(t)^2 |r+1| |r-1|} \\ & + \frac{\frac{\partial}{\partial t} F(t)^2}{8 F(t)^2 |r+1| |r-1|} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1| |r-1|} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} \\ & + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi)} - \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{1}{8 F(t)^2 |r+1| |r-1|} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} \\ & - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8 r^2 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8 r^2 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|}\end{aligned}$$

$$\begin{aligned}\Lambda = & \frac{\frac{\partial}{\partial t} F(t)^2}{12 F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{6 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{6 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{6 F(t)^2 V(t, z, r, \phi)} \\ & + \frac{1}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{12 r F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{12 r^2 F(t)^2 V(t, z, r, \phi)}\end{aligned}$$

Weyl Tensor Components

$$\begin{aligned}
\Psi_0 = & -\frac{r^2 \frac{\partial}{\partial t} F(t)^2}{8(r^2-1)F(t)^2} + \frac{|r+1||r-1| \frac{\partial}{\partial t} F(t)^2}{8(r^2-1)F(t)^2} + \frac{r^2 \frac{\partial^2}{(\partial t)^2} F(t)}{8(r^2-1)F(t)} - \frac{|r+1||r-1| \frac{\partial^2}{(\partial t)^2} F(t)}{8(r^2-1)F(t)} + \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8(r^2-1)F(t)V(t, z, r, \phi)} - \frac{|r+1||r-1| \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8(r^2-1)F(t)V(t, z, r, \phi)} \\
& - \frac{r^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8(r^2-1)V(t, z, r, \phi)} + \frac{|r+1||r-1| \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8(r^2-1)V(t, z, r, \phi)} + \frac{r^2 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2(r^2-1)F(t)V(t, z, r, \phi)} - \frac{r^2 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2(r^2-1)F(t)^2 V(t, z, r, \phi)} + \frac{r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)} \\
& - \frac{r^2 |r+1||r-1| \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)} + \frac{r^3 \frac{\partial}{\partial r} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)} - \frac{r|r+1||r-1| \frac{\partial}{\partial r} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)} - \frac{r^2}{8(r^2-1)F(t)^2} + \frac{|r+1||r-1|}{8(r^2-1)F(t)^2} + \frac{\frac{\partial}{\partial t} F(t)^2}{8(r^2-1)F(t)^2} \\
& - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{8(r^2-1)F(t)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8(r^2-1)F(t)V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8(r^2-1)V(t, z, r, \phi)} - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2(r^2-1)F(t)V(t, z, r, \phi)} + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2(r^2-1)F(t)^2 V(t, z, r, \phi)} \\
& - \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4(r^2-1)F(t)^2 V(t, z, r, \phi)} + \frac{|r+1||r-1| \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)} - \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)} + \frac{1}{8(r^2-1)F(t)^2} + \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8(r^2-1)F(t)^2 V(t, z, r, \phi)}
\end{aligned}$$

$$\begin{aligned}
\Psi_1 = & -\frac{i r^3 \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)V(t, z, r, \phi)} + \frac{i r|r+1||r-1| \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)V(t, z, r, \phi)} - \frac{i r^5 \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i r^3 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} \\
& + \frac{i r^3 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i r|r+1||r-1| \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} + \frac{i r^4 \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{i r^2 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} + \frac{i r \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)V(t, z, r, \phi)} + \frac{i r^3 \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4(r^4-r^2)F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{i r \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i r^2 \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4(r^4-r^2)F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& + \frac{i \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4-r^2)F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}}
\end{aligned}$$

$$\begin{aligned}
\Psi_2 = & \frac{\frac{\partial}{\partial t} F(t)^2}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{12 F(t)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{12 F(t)V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} \\
& - \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} + \frac{1}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{6 r F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{6 r^2 F(t)^2 V(t, z, r, \phi)}
\end{aligned}$$

$$\begin{aligned}
\Psi_3 = & \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 r F(t) V(t, z, r, \phi)} - \frac{i r \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1||r-1|} + \frac{i r^3 \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}}} - \frac{i r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi)} + \frac{i r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1||r-1|} - \frac{i r^2 \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}}} + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& + \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 r F(t) V(t, z, r, \phi) |r+1||r-1|} - \frac{i r \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}}} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi) |r+1||r-1|} \\
& + \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}}} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 r^2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}}} - \frac{i \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 r^2 F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}}} \\
\Psi_4 = & -\frac{\frac{\partial}{\partial t} F(t)^2}{8 F(t)^2} + \frac{r^2 \frac{\partial}{\partial t} F(t)^2}{8 F(t)^2 |r+1||r-1|} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{8 F(t)} - \frac{r^2 \frac{\partial^2}{(\partial t)^2} F(t)}{8 F(t) |r+1||r-1|} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi)} - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1||r-1|} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi)} \\
& + \frac{r^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1||r-1|} + \frac{r^2 \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} - \frac{r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1||r-1|} \\
& + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} - \frac{r^3 \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1||r-1|} - \frac{1}{8 F(t)^2} + \frac{r^2}{8 F(t)^2 |r+1||r-1|} - \frac{\frac{\partial}{\partial t} F(t)^2}{8 F(t)^2 |r+1||r-1|} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{8 F(t) |r+1||r-1|} \\
& + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1||r-1|} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1||r-1|} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) |r+1||r-1|} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1||r-1|} - \frac{1}{8 F(t)^2 |r+1||r-1|} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1||r-1|}
\end{aligned}$$

Newman-Penrose Equations

$$\begin{aligned}
NP2 = & \frac{\frac{\partial}{\partial t} F(t)^2}{8 F(t)^2} + \frac{r^2 \frac{\partial^2}{\partial t^2} F(t)^2}{8 F(t)^2 |r+1| |r-1|} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{8 F(t)} - \frac{r^2 \frac{\partial^2}{(\partial t)^2} F(t)}{8 F(t) |r+1| |r-1|} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi)} - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1| |r-1|} \\
& + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1| |r-1|} + \frac{r^2 \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} \\
& - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} \\
& + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)^2} + \frac{r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2} + \frac{r^4 \frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2 |r+1| |r-1|} - \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} \\
& - \frac{r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} - \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} - \frac{r^3 \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} - \frac{1}{8 F(t)^2} - \frac{r^2}{8 F(t)^2 |r+1| |r-1|} \\
& - \frac{\frac{\partial}{\partial t} F(t)^2}{8 F(t)^2 |r+1| |r-1|} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{8 F(t) |r+1| |r-1|} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1| |r-1|} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1| |r-1|} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2} - \frac{r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)^2}{2 F(t)^2 V(t, z, r, \phi)^2 |r+1| |r-1|} + \dots
\end{aligned}$$

$$NP4 = \frac{i r \frac{\partial}{\partial r} V(t, z, r, \phi) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial r} V(t, z, r, \phi) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)^2} - \frac{i \frac{\partial}{\partial r} V(t, z, r, \phi) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}}$$

9

$$\begin{aligned}
NP7 = & \frac{r^2 \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} - \frac{r^2 \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)^2} + \frac{r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2} + \frac{r^4 \frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2 |r+1| |r-1|} - \frac{1}{4 F(t)^2} \\
& - \frac{r^2}{4 F(t)^2 |r+1| |r-1|} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2} \\
& - \frac{r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)^2}{2 F(t)^2 V(t, z, r, \phi)^2 |r+1| |r-1|} + \frac{1}{4 r^2 F(t)^2} + \frac{1}{2 F(t)^2 |r+1| |r-1|} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)^2}{4 F(t)^2 V(t, z, r, \phi)^2 |r+1| |r-1|} - \frac{1}{4 r^2 F(t)^2 |r+1| |r-1|}
\end{aligned}$$

10

$$\begin{aligned}
NP12 = & \frac{r^2 \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi)} - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} \\
& + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi)} + \frac{|r+1| |r-1| \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 r F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{2 r F(t)^2 V(t, z, r, \phi)}
\end{aligned}$$

$$\begin{aligned}
NP13 = & \frac{i r^3 \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t) V(t, z, r, \phi)} + \frac{i r |r+1| |r-1| \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t) V(t, z, r, \phi)} - \frac{i r^3 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} - \frac{i r |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}} \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} \\
& - \frac{i r^3 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} - \frac{i r |r+1| |r-1| \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} + \frac{i r^2 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} + \frac{i |r+1|^{\frac{3}{2}} |r-1|^{\frac{3}{2}} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} \\
& - \frac{i r \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t) V(t, z, r, \phi)} + \frac{i r \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} + \frac{i r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)} - \frac{i \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8 (r^4 - r^2) F(t)^2 V(t, z, r, \phi)}
\end{aligned}$$

$$\begin{aligned}
NP14 = & \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi)} + \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1| |r-1|} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1| |r-1|} - \frac{r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} + \frac{r^4 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} - \frac{r \frac{\partial}{\partial t} F(t)}{8 F(t)^2 \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t)}{8 r F(t)^2} + \frac{r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{r^3 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{1}{8 F(t)^2} + \frac{r^2}{8 F(t)^2 |r+1| |r-1|} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{8 F(t) V(t, z, r, \phi) |r+1| |r-1|} \\
& - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{8 V(t, z, r, \phi) |r+1| |r-1|} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} - \frac{r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} \\
& + \frac{\frac{\partial}{\partial t} F(t)}{8 r F(t)^2 \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi)} - \frac{3 r \frac{\partial}{\partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} - \frac{1}{8 F(t)^2 |r+1| |r-1|} + \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} \\
& - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8 r^2 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{8 r F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{8 r^2 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|}
\end{aligned}$$

$$\begin{aligned}
NP16 = & -\frac{\partial F(t)^2}{8F(t)^2} - \frac{r^2 \frac{\partial F(t)^2}{\partial t}}{8F(t)^2 |r+1||r-1|} + \frac{\frac{\partial F(t)}{\partial t} \frac{\partial V(t, z, r, \phi)}{\partial t}}{8F(t) V(t, z, r, \phi)} + \frac{r^2 \frac{\partial F(t)}{\partial t} \frac{\partial V(t, z, r, \phi)}{\partial t}}{8F(t) V(t, z, r, \phi) |r+1||r-1|} + \frac{r^2 \frac{\partial F(t)}{\partial t} \frac{\partial V(t, z, r, \phi)}{\partial r}}{8F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& + \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial F(t)}{\partial t} \frac{\partial V(t, z, r, \phi)}{\partial r}}{8F(t)^2 V(t, z, r, \phi)} + \frac{r^2 \frac{\partial^2 V(t, z, r, \phi)}{(\partial r)^2}}{4F(t)^2 V(t, z, r, \phi)} + \frac{r^4 \frac{\partial^2 V(t, z, r, \phi)}{(\partial r)^2}}{4F(t)^2 V(t, z, r, \phi) |r+1||r-1|} - \frac{r \frac{\partial F(t)}{\partial t}}{8F(t)^2 \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{\sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial F(t)}{\partial t}}{8rF(t)^2} + \frac{r \frac{\partial V(t, z, r, \phi)}{\partial r}}{4F(t)^2 V(t, z, r, \phi)} + \frac{r^3 \frac{\partial V(t, z, r, \phi)}{\partial r}}{4F(t)^2 V(t, z, r, \phi) |r+1||r-1|} - \frac{1}{4F(t)^2} - \frac{r^2}{4F(t)^2 |r+1||r-1|} + \frac{\frac{\partial F(t)^2}{\partial t}}{8F(t)^2 |r+1||r-1|} \\
& - \frac{\frac{\partial F(t)}{\partial t} \frac{\partial V(t, z, r, \phi)}{\partial t}}{8F(t) V(t, z, r, \phi) |r+1||r-1|} - \frac{\frac{\partial F(t)}{\partial t} \frac{\partial V(t, z, r, \phi)}{\partial r}}{8F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{\frac{\partial^2 V(t, z, r, \phi)}{(\partial r)^2}}{4F(t)^2 V(t, z, r, \phi)} - \frac{r^2 \frac{\partial^2 V(t, z, r, \phi)}{(\partial r)^2}}{2F(t)^2 V(t, z, r, \phi) |r+1||r-1|} \\
& + \frac{\frac{\partial F(t)}{\partial t}}{8rF(t)^2 \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{r \frac{\partial V(t, z, r, \phi)}{\partial r}}{4F(t)^2 V(t, z, r, \phi) |r+1||r-1|} + \frac{1}{4F(t)^2 |r+1||r-1|} + \frac{\frac{\partial^2 V(t, z, r, \phi)}{(\partial r)^2}}{4F(t)^2 V(t, z, r, \phi) |r+1||r-1|}
\end{aligned}$$

$$NP18 = \frac{i r^3 \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)V(t, z, r, \phi)} + \frac{i r|r+1||r-1| \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)V(t, z, r, \phi)} - \frac{i r^3 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i r|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)}$$

$$- \frac{i r^3 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i r|r+1||r-1| \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{i r^2 \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{i|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)}$$

$$- \frac{i r \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)V(t, z, r, \phi)} + \frac{i r \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} + \frac{i r \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)} - \frac{i \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{8(r^4 - r^2)F(t)^2 V(t, z, r, \phi)}$$

Bianchi Identities

$$\begin{aligned}
BI1 = & \frac{\sqrt{2} \frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial t \partial z} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^2} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial t \partial z} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^2 |r+1| |r-1|} - \frac{\sqrt{2} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial z} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^3} - \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial}{\partial z} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^3 |r+1| |r-1|} \\
& + \frac{i \sqrt{2} \frac{\partial^3}{(\partial t)^2 \partial \phi} V(t, z, r, \phi)}{16 r F(t) V(t, z, r, \phi)} + \frac{i \sqrt{2} r \frac{\partial^3}{(\partial t)^2 \partial \phi} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi) |r+1| |r-1|} + \frac{i \sqrt{2} r \frac{\partial^3}{\partial t \partial r \partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \sqrt{2} r \sqrt{|r-1|} \frac{\partial^3}{\partial t \partial r \partial \phi} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) |r+1|^{\frac{3}{2}}} \\
& - \frac{i \sqrt{2} \frac{\partial}{\partial z} V(t, z, r, \phi) \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{16 r F(t) V(t, z, r, \phi)^2} - \frac{i \sqrt{2} r \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^2 |r+1| |r-1|} + \frac{\sqrt{2} r \frac{\partial^2}{\partial z \partial r} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^2} + \frac{\sqrt{2} r^3 \frac{\partial^2}{\partial z \partial r} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^2 |r+1| |r-1|} \\
& + \frac{i \sqrt{2} r \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \sqrt{2} r \sqrt{|r-1|} \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^2 |r+1|^{\frac{3}{2}}} - \frac{\sqrt{2} r \frac{\partial}{\partial z} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^3} \\
& - \frac{\sqrt{2} r^3 \frac{\partial}{\partial z} V(t, z, r, \phi) \frac{\partial}{\partial r} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^3 |r+1| |r-1|} + \frac{i \sqrt{2} r \frac{\partial^3}{(\partial r)^2 \partial \phi} V(t, z, r, \phi)}{16 F(t)^3 V(t, z, r, \phi)} + \frac{i \sqrt{2} r^3 \frac{\partial^3}{(\partial r)^2 \partial \phi} V(t, z, r, \phi)}{16 F(t)^3 V(t, z, r, \phi) |r+1| |r-1|} - \frac{i \sqrt{2} r \frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8 F(t)^3 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} \\
& - \frac{i \sqrt{2} r \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{8 F(t)^3 V(t, z, r, \phi) |r+1|^{\frac{3}{2}}} + \frac{i \sqrt{2} r \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{i \sqrt{2} r \sqrt{|r-1|} \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)^2 |r+1|^{\frac{3}{2}}} - \dots
\end{aligned}$$

$$\begin{aligned}
BI2 = & \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)^3}{16 F(t)^3} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} F(t)^3}{16 F(t)^3 |r+1| |r-1|} + \frac{\sqrt{2} \frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} F(t)}{32 F(t)^2} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} F(t)}{32 F(t)^2 |r+1| |r-1|} - \frac{\sqrt{2} \frac{\partial^3}{(\partial t)^3} F(t)}{16 F(t)} - \frac{\sqrt{2} r^2 \frac{\partial^3}{(\partial t)^3} F(t)}{16 F(t) |r+1| |r-1|} \\
& - \frac{3 \sqrt{2} \frac{\partial}{\partial t} F(t)^2 \frac{\partial}{\partial t} V(t, z, r, \phi)}{32 F(t)^2 V(t, z, r, \phi)} - \frac{3 \sqrt{2} r^2 \frac{\partial}{\partial t} F(t)^2 \frac{\partial}{\partial t} V(t, z, r, \phi)}{32 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} - \frac{3 \sqrt{2} \frac{\partial^2}{(\partial t)^2} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{32 F(t) V(t, z, r, \phi)} - \frac{3 \sqrt{2} r^2 \frac{\partial^2}{(\partial t)^2} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{32 F(t) V(t, z, r, \phi) |r+1| |r-1|} \\
& + \frac{\sqrt{2} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)^2}{32 F(t) V(t, z, r, \phi)^2} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)^2}{32 F(t) V(t, z, r, \phi)^2 |r+1| |r-1|} + \frac{\sqrt{2} \frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi) |r+1| |r-1|} \\
& + \frac{\sqrt{2} \frac{\partial^3}{(\partial t)^3} V(t, z, r, \phi)}{16 V(t, z, r, \phi)} + \frac{\sqrt{2} r^2 \frac{\partial^3}{(\partial t)^3} V(t, z, r, \phi)}{16 V(t, z, r, \phi) |r+1| |r-1|} + \frac{\sqrt{2} r^2 \frac{\partial^3}{(\partial t)^2 \partial r} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \frac{\sqrt{2} \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial^3}{(\partial t)^2 \partial r} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)} \\
& - \frac{3 \sqrt{2} r^2 \frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} - \frac{3 \sqrt{2} \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8 F(t)^2 V(t, z, r, \phi)} + \frac{\sqrt{2} r^2 \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^2 \sqrt{|r+1|} \sqrt{|r-1|}} \\
& + \frac{\sqrt{2} \sqrt{|r+1|} \sqrt{|r-1|} \frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{16 F(t) V(t, z, r, \phi)^2} - \frac{\sqrt{2} r^2 \frac{\partial^3}{\partial t (\partial r)^2} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi)} - \frac{\sqrt{2} r^4 \frac{\partial^3}{\partial t (\partial r)^2} V(t, z, r, \phi)}{16 F(t)^2 V(t, z, r, \phi) |r+1| |r-1|} + \frac{5 \sqrt{2} r^2 \frac{\partial}{\partial t} F(t)^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{16 F(t)^3 V(t, z, r, \phi) \sqrt{|r+1|} \sqrt{|r-1|}} + \dots
\end{aligned}$$

$$\begin{aligned}
BI3 = & -\frac{\sqrt{2}r^5 \frac{\partial}{\partial t} F(t)^3}{16(r^5 - r^3)F(t)^3} - \frac{\sqrt{2}r^3|r+1||r-1|\frac{\partial}{\partial t} F(t)^3}{16(r^5 - r^3)F(t)^3} - \frac{\sqrt{2}r^5 \frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} F(t)}{32(r^5 - r^3)F(t)^2} - \frac{\sqrt{2}r^3|r+1||r-1|\frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} F(t)}{32(r^5 - r^3)F(t)^2} + \frac{\sqrt{2}r^5 \frac{\partial^3}{(\partial t)^3} F(t)}{16(r^5 - r^3)F(t)} \\
& + \frac{\sqrt{2}r^3|r+1||r-1|\frac{\partial^3}{(\partial t)^3} F(t)}{16(r^5 - r^3)F(t)} + \frac{3\sqrt{2}r^5 \frac{\partial}{\partial t} F(t)^2 \frac{\partial}{\partial t} V(t, z, r, \phi)}{32(r^5 - r^3)F(t)^2 V(t, z, r, \phi)} + \frac{3\sqrt{2}r^3|r+1||r-1|\frac{\partial}{\partial t} F(t)^2 \frac{\partial}{\partial t} V(t, z, r, \phi)}{32(r^5 - r^3)F(t)^2 V(t, z, r, \phi)} + \frac{3\sqrt{2}r^5 \frac{\partial^2}{(\partial t)^2} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{32(r^5 - r^3)F(t) V(t, z, r, \phi)} \\
& + \frac{3\sqrt{2}r^3|r+1||r-1|\frac{\partial^2}{(\partial t)^2} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{32(r^5 - r^3)F(t) V(t, z, r, \phi)} - \frac{\sqrt{2}r^5 \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)^2}{32(r^5 - r^3)F(t) V(t, z, r, \phi)^2} - \frac{\sqrt{2}r^3|r+1||r-1|\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)^2}{32(r^5 - r^3)F(t) V(t, z, r, \phi)^2} \\
& - \frac{\sqrt{2}r^5 \frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{16(r^5 - r^3)F(t) V(t, z, r, \phi)} - \frac{\sqrt{2}r^3|r+1||r-1|\frac{\partial}{\partial t} F(t) \frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{16(r^5 - r^3)F(t) V(t, z, r, \phi)} - \frac{\sqrt{2}r^5 \frac{\partial^3}{(\partial t)^3} V(t, z, r, \phi)}{16(r^5 - r^3)V(t, z, r, \phi)} - \frac{\sqrt{2}r^3|r+1||r-1|\frac{\partial^3}{(\partial t)^3} V(t, z, r, \phi)}{16(r^5 - r^3)V(t, z, r, \phi)} \\
& - \frac{\sqrt{2}r^5 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{(\partial t)^2 \partial r} V(t, z, r, \phi)}{16(r^5 - r^3)F(t) V(t, z, r, \phi)} - \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^3}{(\partial t)^2 \partial r} V(t, z, r, \phi)}{16(r^5 - r^3)F(t) V(t, z, r, \phi)} + \frac{\sqrt{2}r^5 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8(r^5 - r^3)F(t)^2 V(t, z, r, \phi)} \\
& + \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t} F(t) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{8(r^5 - r^3)F(t)^2 V(t, z, r, \phi)} - \frac{\sqrt{2}r^5 \sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{16(r^5 - r^3)F(t) V(t, z, r, \phi)^2} - \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t} V(t, z, r, \phi) \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{16(r^5 - r^3)F(t) V(t, z, r, \phi)^2} + \dots
\end{aligned}$$

$$\begin{aligned}
BI4 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial z}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial z}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2|r+1||r-1|} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^3} \\
& + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^3|r+1||r-1|} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial\phi}V(t,z,r,\phi)}{16rF(t)V(t,z,r,\phi)} + \frac{i\sqrt{2}r\frac{\partial^3}{(\partial t)^2\partial\phi}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)|r+1||r-1|} + \frac{i\sqrt{2}r^3\frac{\partial^3}{\partial t\partial r\partial\phi}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}} \\
& + \frac{i\sqrt{2}r\frac{\partial^3}{\partial t\partial r\partial\phi}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{16rF(t)^2V(t,z,r,\phi)} + \frac{i\sqrt{2}r\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)|r+1||r-1|} - \frac{i\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{16rF(t)V(t,z,r,\phi)^2} \\
& - \frac{i\sqrt{2}r\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2|r+1||r-1|} - \frac{\sqrt{2}r\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2} - \frac{\sqrt{2}r^3\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2|r+1||r-1|} + \frac{i\sqrt{2}r^3\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}} \\
& + \frac{i\sqrt{2}r\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}r\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^3} + \frac{\sqrt{2}r^3\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^3|r+1||r-1|} + \frac{i\sqrt{2}r\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}r^3\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)|r+1||r-1|} - \frac{3i\sqrt{2}r^3\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}} - \frac{3i\sqrt{2}r\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \dots
\end{aligned}$$

$$\begin{aligned}
BI9 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{16F(t)^3} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)^3}{16F(t)^3|r+1||r-1|} - \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)} - \frac{3\sqrt{2}r^2\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)|r+1||r-1|} + \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)} \\
& + \frac{3\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)|r+1||r-1|} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{16V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{16V(t,z,r,\phi)|r+1||r-1|} + \frac{\sqrt{2}r^2\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{8F(t)V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} \\
& + \frac{\sqrt{2}r^2\sqrt{|r-1|}\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{8F(t)V(t,z,r,\phi)|r+1|^{\frac{3}{2}}} - \frac{3\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{3\sqrt{2}r^2\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)|r+1|^{\frac{3}{2}}} \\
& + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}r^2\sqrt{|r-1|}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2|r+1|^{\frac{3}{2}}} - \frac{\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}r^4\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)|r+1||r-1|} \\
& + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}r^2\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)|r+1|^{\frac{3}{2}}} - \frac{5\sqrt{2}r^2\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} \\
& - \frac{5\sqrt{2}r^2\sqrt{|r-1|}\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)|r+1|^{\frac{3}{2}}} + \frac{5\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{5\sqrt{2}r^2\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2|r+1|^{\frac{3}{2}}} + \dots
\end{aligned}$$

$$\begin{aligned}
BI10 = & \frac{\sqrt{2}r^5\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{\partial t\partial z\partial r}V(t,z,r,\phi)}{16(r^5-r^3)F(t)V(t,z,r,\phi)^2} + \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^3}{\partial t\partial z\partial r}V(t,z,r,\phi)}{16(r^5-r^3)F(t)V(t,z,r,\phi)^2} - \frac{\sqrt{2}r^5\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{16(r^5-r^3)F(t)V(t,z,r,\phi)^3} \\
& - \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{16(r^5-r^3)F(t)V(t,z,r,\phi)^3} - \frac{\sqrt{2}r^5\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)^2} - \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}r^5\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)^3} + \frac{\sqrt{2}r^3|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)^3} \\
& - \frac{3i\sqrt{2}r^4\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{\partial t\partial r\partial\phi}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)} - \frac{3i\sqrt{2}r^2|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^3}{\partial t\partial r\partial\phi}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)} - \frac{i\sqrt{2}r^4\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}r^2|r+1||r-1|\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^2V(t,z,r,\phi)} + \frac{i\sqrt{2}r^4\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8(r^5-r^3)F(t)^2V(t,z,r,\phi)^2} + \frac{i\sqrt{2}r^2|r+1|^{\frac{3}{2}}|r-1|^{\frac{3}{2}}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8(r^5-r^3)F(t)^2V(t,z,r,\phi)^2} \\
& - \frac{i\sqrt{2}r^6\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{8(r^5-r^3)F(t)^3V(t,z,r,\phi)^2} - \frac{i\sqrt{2}r^4|r+1||r-1|\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{8(r^5-r^3)F(t)^3V(t,z,r,\phi)^2} + \frac{i\sqrt{2}r^4\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{16(r^5-r^3)F(t)^3V(t,z,r,\phi)} + \dots
\end{aligned}$$

$$\begin{aligned}
BI11 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{16F(t)^3} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)^3}{16F(t)^3|r+1||r-1|} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}F(t)}{8F(t)^2} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}F(t)}{8F(t)^2|r+1||r-1|} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)|r+1||r-1|} \\
& + \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)} + \frac{3\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)|r+1||r-1|} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{16V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{16V(t,z,r,\phi)|r+1||r-1|} \\
& + \frac{\sqrt{2}r^2\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{8F(t)V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{8F(t)V(t,z,r,\phi)} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} \\
& - \frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}} + \frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2} \\
& - \frac{\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}r^4\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)|r+1||r-1|} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}r^2\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)\sqrt{|r+1|}\sqrt{|r-1|}} - \frac{\sqrt{2}\sqrt{|r+1|}\sqrt{|r-1|}\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2\sqrt{|r+1|}\sqrt{|r-1|}} + \dots
\end{aligned}$$

Equations are not vanished.

Petrov Type

Type "I"